REES46 | Customer conversion measurement

Observed IDs and reconstructed 30-minute sessions. Historical snapshot; descriptive, not a causal effect.

Retained Events	Observed Users	Observed Products	Observed Sessions
884K	**407K**	**53K**	**505K**

Daily events | UTC, 24 Sep 2020 - 28 Feb 2021



Metric definitions and data quality

Rates use different units and windows. Quality checks overlap; do not add affected rows.

MetricID	MetricName	Unit	Snapshot Rate
F01	Within-session product view-to-cart progression	user-session-product	7.67%
F02	Within-session ordered product purchase progression	user-session-product	3.67%
F03	Purchase after cart among view-entered product groups	user-session-product	47.86%
P01	7-day observed view-to-cart progression	user-product pair	7.98%
P02	7-day ordered view-cart-purchase progression	user-product pair	3.99%
P03	7-day observed cart-to-purchase progression	user-product pair	49.96%
S01	30-minute session purchase share	reconstructed user-session	4.86%
U01	Observed-user purchase share	observed user	5.23%

Check	Sum of AffectedRows	Sum of BaseRows	Quality Affected Share
Exact duplicate extra raw rows	655	885129	0.07%
Missing brand in retained events	212232	884474	24.00%
Missing category in retained events	236047	884474	26.69%
Missing supplied session in retained events	162	884474	0.02%

Category funnel | Closed 30-minute sessions, first-view category

Strict view < cart < purchase for the same product. Exploratory segments; small denominators need care.

Category Eligible Groups	Category Carts	Category Ordered Purchases	Category Ordered Purchase Rate
645K	49K	24K	3.67%

Category

Search

- [ ] accessories.bag
- [ ] accessories.briefcase
- [ ] apparel.glove
- [ ] appliances.environment.air_heater
- [ ] appliances.environment.fan
- [ ] appliances.environment.vacuum
- [ ] appliances.iron
- [ ] appliances.kitchen.blender
- [ ] appliances.kitchen.coffee_grinder
- [ ] appliances.kitchen.coffee_machine

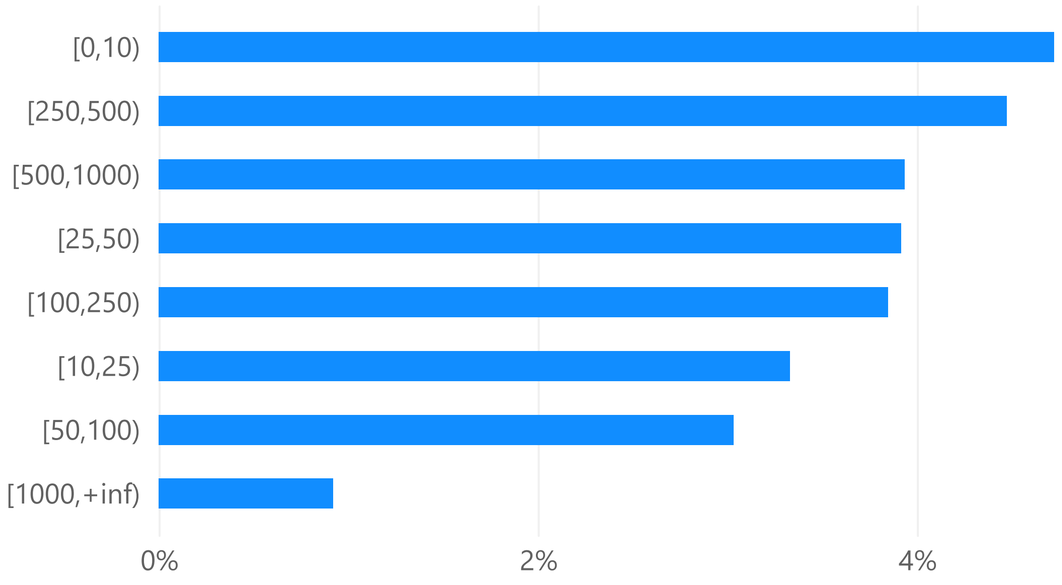
Category	Category Eligible Groups	Category Ordered Purchases	Category Ordered Purchase Rate
Unknown	180,978	4,778	2.64%
computers.components.videocards	72,098	4,712	6.54%
electronics.telephone	61,107	2,652	4.34%
computers.peripherals.printer	30,000	1,579	5.26%
stationery.cartrige	27,113	1,679	6.19%
electronics.audio.acoustic	20,447	486	2.38%
computers.components.motherboard	18,540	778	4.20%
computers.notebook	18,012	818	4.54%
electronics.video.tv	17,065	228	1.34%
computers.components.cpu	16,532	730	4.42%
electronics.tablet	14,192	524	3.69%
Total	**644,660**	**23,672**	**3.67%**

Price funnel | First-view price band, closed 30-minute sessions
Source price units; currency is unspecified. Click a bar to explore. Price mix is observational, not a price effect.

Price Eligible Groups	Price Carts	Price Ordered Purchases	Price Ordered Purchase Rate
645K	49K	24K	3.67%

Price Ordered Purchase Rate

by PriceBand

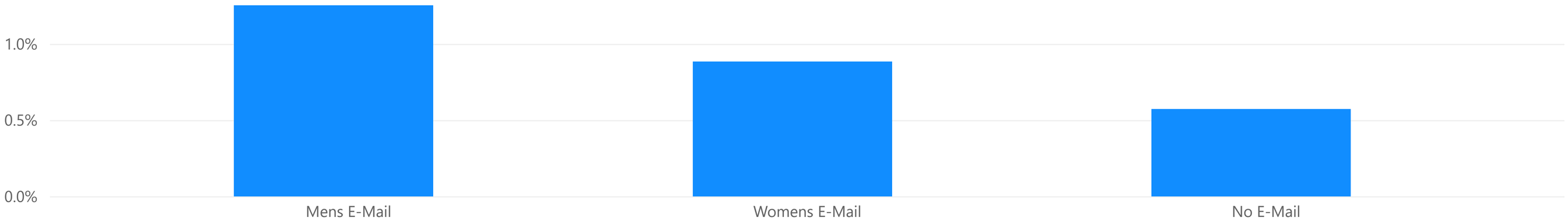


PriceBand	Price Eligible Groups	Price Ordered Purchases	Price Ordered Purchase Rate
[0,10)	19,694	930	4.72%
[10,25)	140,471	4,676	3.33%
[100,250)	128,326	4,938	3.85%
[1000,+inf)	5,443	50	0.92%
[25,50)	115,609	4,528	3.92%
[250,500)	83,230	3,724	4.47%
[50,100)	127,540	3,868	3.03%
[500,1000)	24,347	958	3.93%
Total	**644,660**	**23,672**	**3.67%**

Hillstrom 2008 | Separate randomized email experiment
64,000 assigned customers; all rows and zero spend retained. Historical dollars are not current profit or ROI.

Experimental Purchase Rate

by Arm



Arm	Assigned Customers	Experimental Purchase Rate	Experimental Visit Rate	Mean Recorded Spend
Mens E-Mail	21,307	1.25%	18.28%	$1.4226
No E-Mail	21,306	0.57%	10.62%	$0.6528
Womens E-Mail	21,387	0.88%	15.14%	$1.0772

Hillstrom | Treatment minus control

Primary purchase: percentage points, Holm tests. 95% CIs are marginal. Secondary spend: dollars per assigned customer.

Treatment	Purchase Difference pp	Purchase Lower pp	Purchase Upper pp	Holm Adjusted P
Mens E-Mail	0.6805	0.5014	0.8641	3.05E-13
Womens E-Mail	0.3111	0.1502	0.4745	1.57E-04

Treatment	Spend Difference Dollars	Spend Lower Dollars	Spend Upper Dollars
Mens E-Mail	$0.7698	$0.4976	$1.0566
Womens E-Mail	$0.4244	$0.1703	$0.6828